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TECHNICAL SUMMARY

Load Flow Analysis

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ABSTRACT

Newton–Raphson load flow in Python and ETAP for multi-bus distribution systems. Developed and implemented Newton–Raphson load-flow algorithms in Python and ETAP for 10, 20 and 30-bus distribution systems, and cross-validated the two.

Bus systems	10 / 20 / 30
Solver	Newton–Raphson

1. Problem and motivation

Distribution engineers need bus voltages and losses along a feeder to plan upgrades. Long rural feeders often end with voltages below the acceptable limit.

2. Methodology

- Implemented Newton–Raphson load flow in Python and ETAP
- Analyzed voltage profiles, real and reactive power flow, and system losses
- Evaluated performance under varying load, identifying voltage stability limits and how losses are distributed
- Cross-validated results between Python and ETAP

3. Figures

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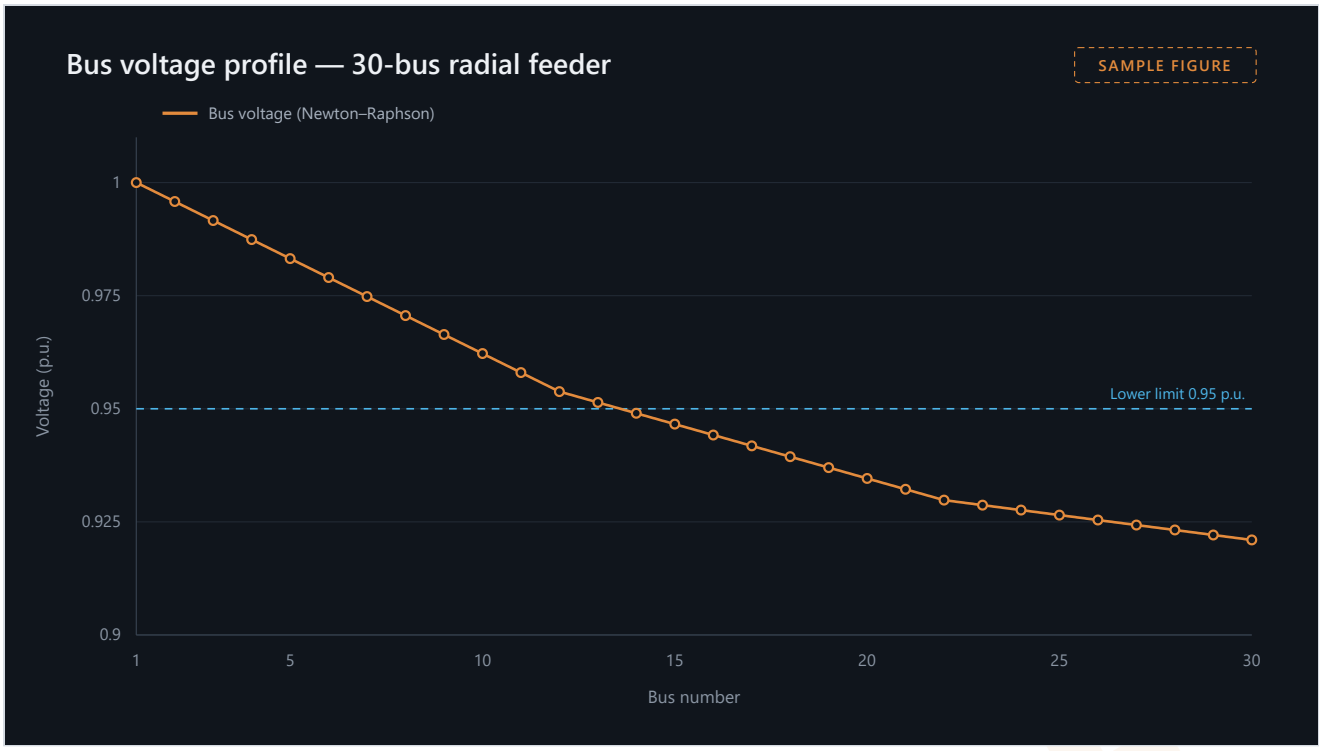


Figure 1. Bus voltage profile along the 30-bus feeder.

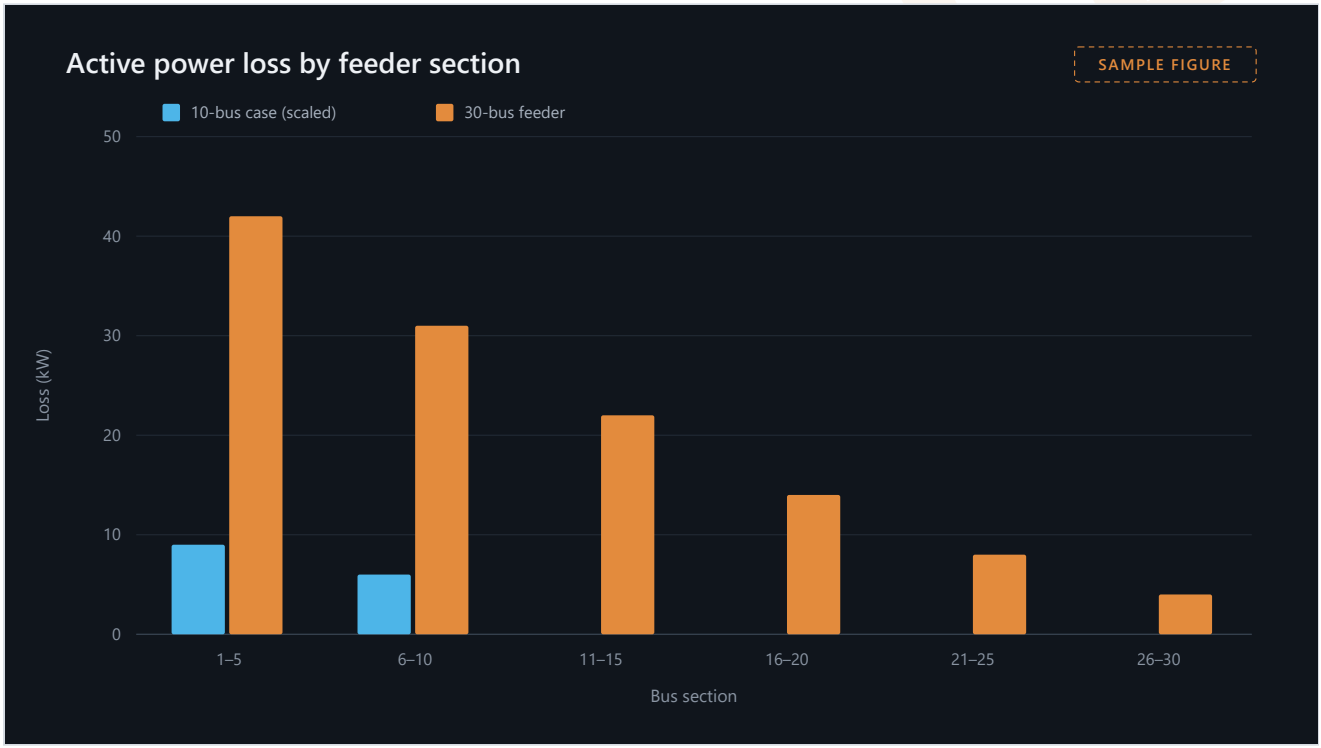


Figure 2. Active power loss by feeder section.

4. Team and contribution

Contribution: Wrote the Newton–Raphson solver in Python and validated it against ETAP.

Tools: Python, ETAP, Power system analysis

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